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Unit 3 - Creating Your Own Mood Light

PDF Documents

PDF Document 1: 1737526637160-Chapter 5 - unit 3 - Creating Your Own Mood Light.pdf

This PDF document is included in the unit materials.



UNIT 3 CREATING YOUR OWN MOOD LIGHT

Today, we are going to create a fun Mood Lantern! This lantern changes colors every time you press a button. Let's see how it works:



HOW IT WORKS IN ACTION:

1 Starting with Mode 0 (Off)

When you turn on the lantern, it starts in Mode 0. This means that the light is off, and there is no color yet.

2 Pressing the Button

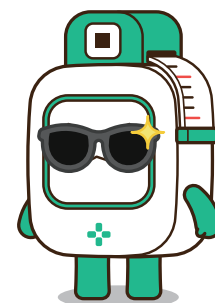
When you press the button, the mode changes. Each mode will make the lantern light up in a different color.

- **Mode 1:** When the mode is 1, the lantern glows yellow.
- **Mode 2:** When the mode is 2, the lantern changes to green.

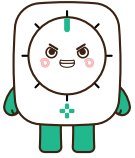
3 Resetting the Mode

If you press the button again and the mode goes above 2, it will reset back to Mode 0. This means the light will turn off again.

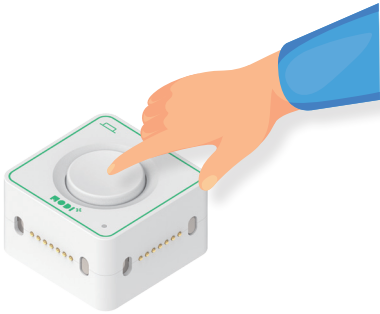
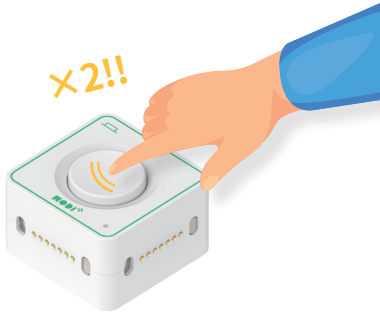
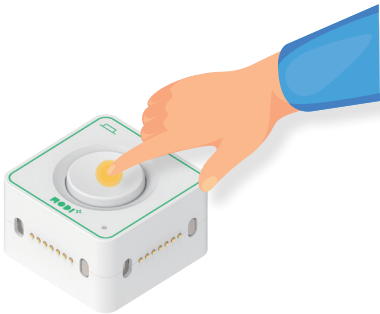
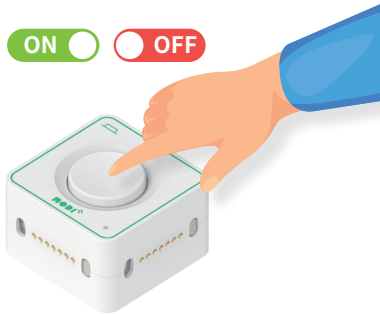
Every time you press the button, the lantern switches between being off, yellow, and green, depending on the mode. This way, you can easily control the mood in the room with just a few button presses!



Do you remember which action Button can received?



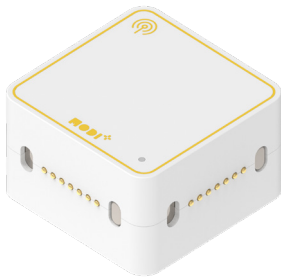
Button can receive 4 types of information.

1. CLICK	2. DOUBLE CLICK
	
Detects if Button is clicked or not.	Detects if Button is double - clicked or not.
3. PRESS	4. TOGGLE
	
Show the press status of Button.	Changes on / off when Button is clicked.



Activity: Create Restroom Indicator

1 Materials Needed:



Network



Button



LED

2 Code story

When Button is Clicked



add mode 1



if mode = 0

mode = 1

mode = 2



turn off LED

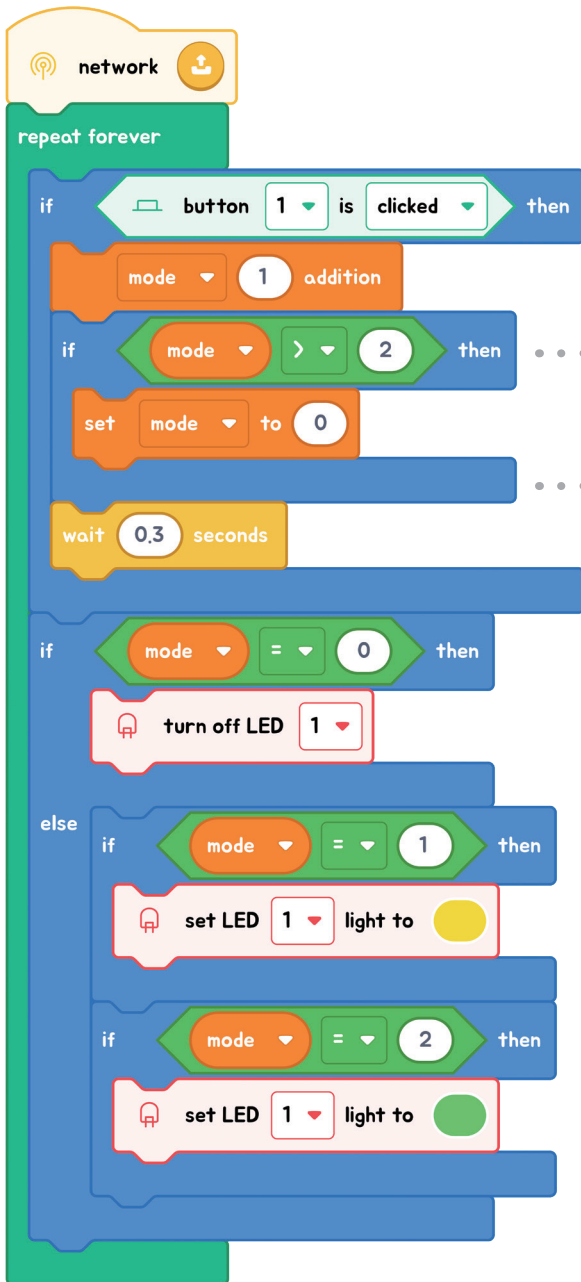


LED as yellow

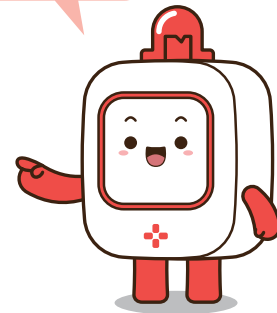


LED as green

Code Quest



When the mode is over than 2, set the mode to 0 because if the mode is 3 there isn't option to choose the light color.



Checkpoint

If you want to add different colors to the light, which block would you need to add?